

# EUROGEOSS WORKSHOP 2018

#EGW2018

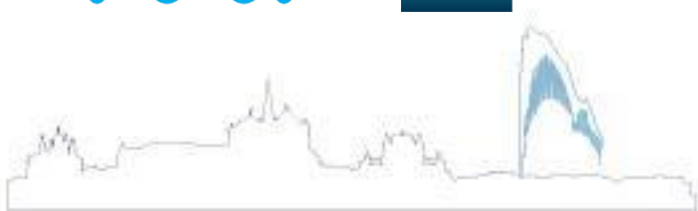
12-14 SEPTEMBER 2018, GENEVA





Pier Luigi Buttigieg (AWI)

# Hands on with the UNESCO/IOC-IODE Ocean Best Practices System



# The core Best Practices Working Group

- Mark Bushnell IOOS
- Pier Luigi Buttigieg AWI
- Juliet Hermes SAEON/JCOMM
- Emma Heslop SOCIB/JCOMM
- Frank Muller-Karger MBON/U South Florida
- Cristian Muñoz SOCIB
- Francoise Pearlman IEEE
- Jay Pearlman IEEE
- Pauline Simpson IODE



# Objectives

Through a use case centered on the UNESCO/IOC-IODE Ocean Best Practices repository, participants will be shown how machine-readable expert knowledge representations (i.e. ontologies) can enhance access to and discovery of quality methods for marine observation and place them in relation to the SDGs.

- [www.oceanbestpractices.org](http://www.oceanbestpractices.org) (last release)
- <https://tinyurl.com/obpBeta> (next release)
- <https://www.oceanbestpractices.net/feedback> (your feedback)



# Objectives

**Participant feedback will be used to enhance the system and improve its fitness for their own activities.**

In your opinion, which marine phenomena should be linked to SDG monitoring, extending the official indicators?

<https://tinyurl.com/EuroGEOSS-sdgio-input>





## **The main goal:**

Transition from a loosely-coordinated set of existing ocean observing activities to a sustainable, efficient, and fit-for-purpose Integrated Atlantic Ocean Observing System (IAOOS)

[www.atlantos-h2020.eu](http://www.atlantos-h2020.eu)

#EGW2018





## WP6 Cross-cutting issues and emerging networks

[...]

Coordinate engagement of international expertise and dissemination of best practice to improve efficiency and impact

[...]

to include international expertise in the design and development, operation and maintenance of IAOS and to disseminate this best practice



# Best Practice

A documented procedure which, through experience and research, has consistently shown results superior to those achieved by other means and can be used as a benchmark, particularly if advocacy can lead to it being widely adopted

*...modulo cost, feasibility, and  
context  
...and which has a process for  
evolution*

**BPs should be catalysts for “Next Practices”**

*Mike Myatt,  
Best Practices - Aren't,  
Forbes. AUG 15, 2012 @  
04:16 PM*

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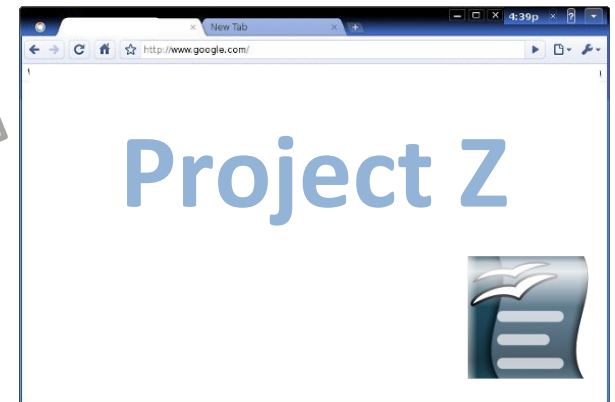
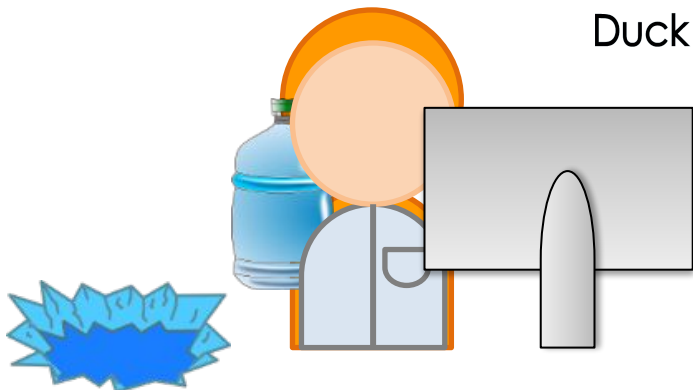
**Most potential BPs are scattered, have varying degrees of accessibility, and varying digital lifetimes**

**404**

**Not found**



DuckDuckGo



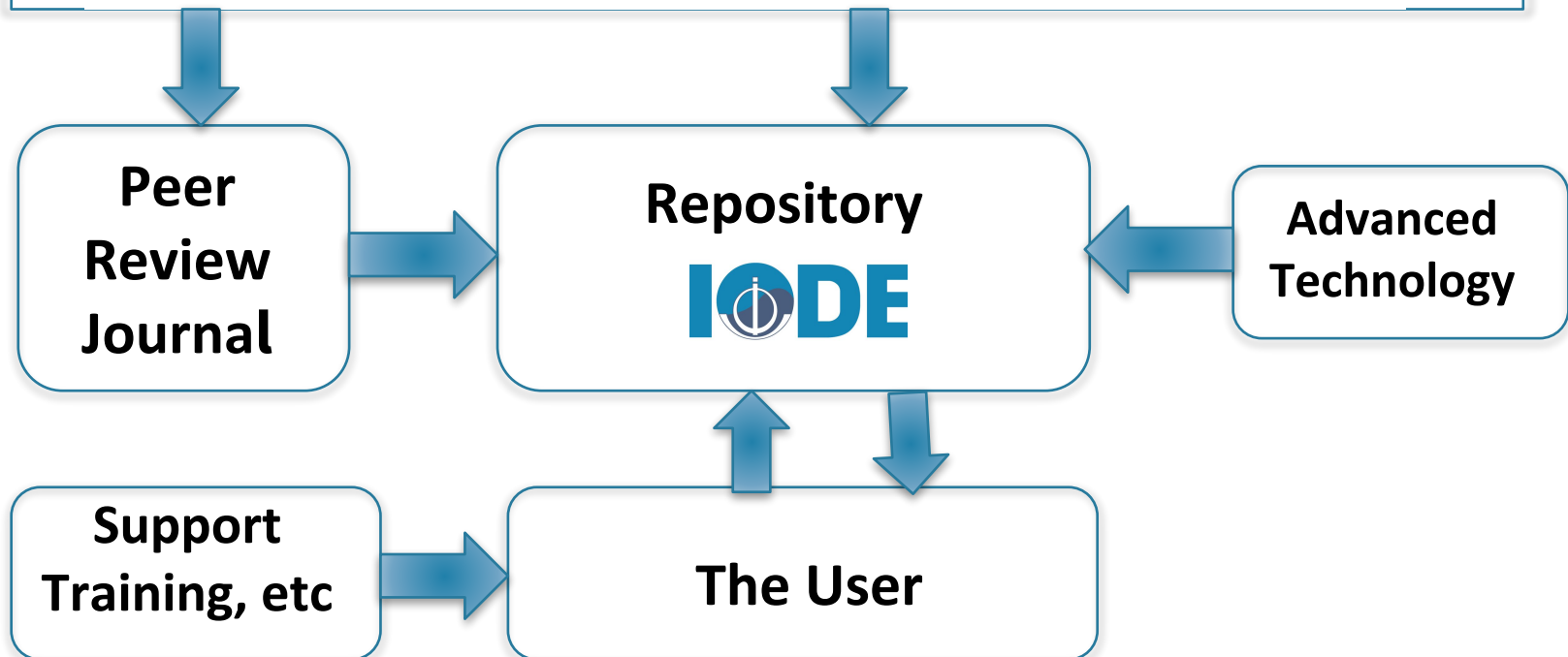
# The Mission: solutions for a FAIRer future for Ocean Best Practices

**FAIR: Findable, Accessible, Interoperable, Reusable**

Wilkinson et al. (2016) *Scientific Data* 3, DOI:10.1038/sdata.2016.18

# Global Best Practice System

## Participating Organizations and Programs





Disclaimer: IODE/IOC does not warrant that the information, documents and materials contained in the OceanBestPractices repository website is complete and correct and shall not be liable whatsoever for any damages incurred as a result of its use.

BROWSE

All of OceanBestPractices

Communities & Collections

By Issue Date

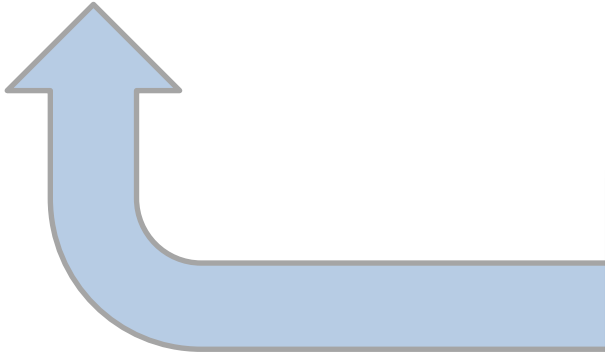
Authors

Titles

Subjects

# OceanBestPractices

Repository of Ocean Community Practices in Ocean Research, Observation and Data/Information Management



My BP

Author Author

Part 1

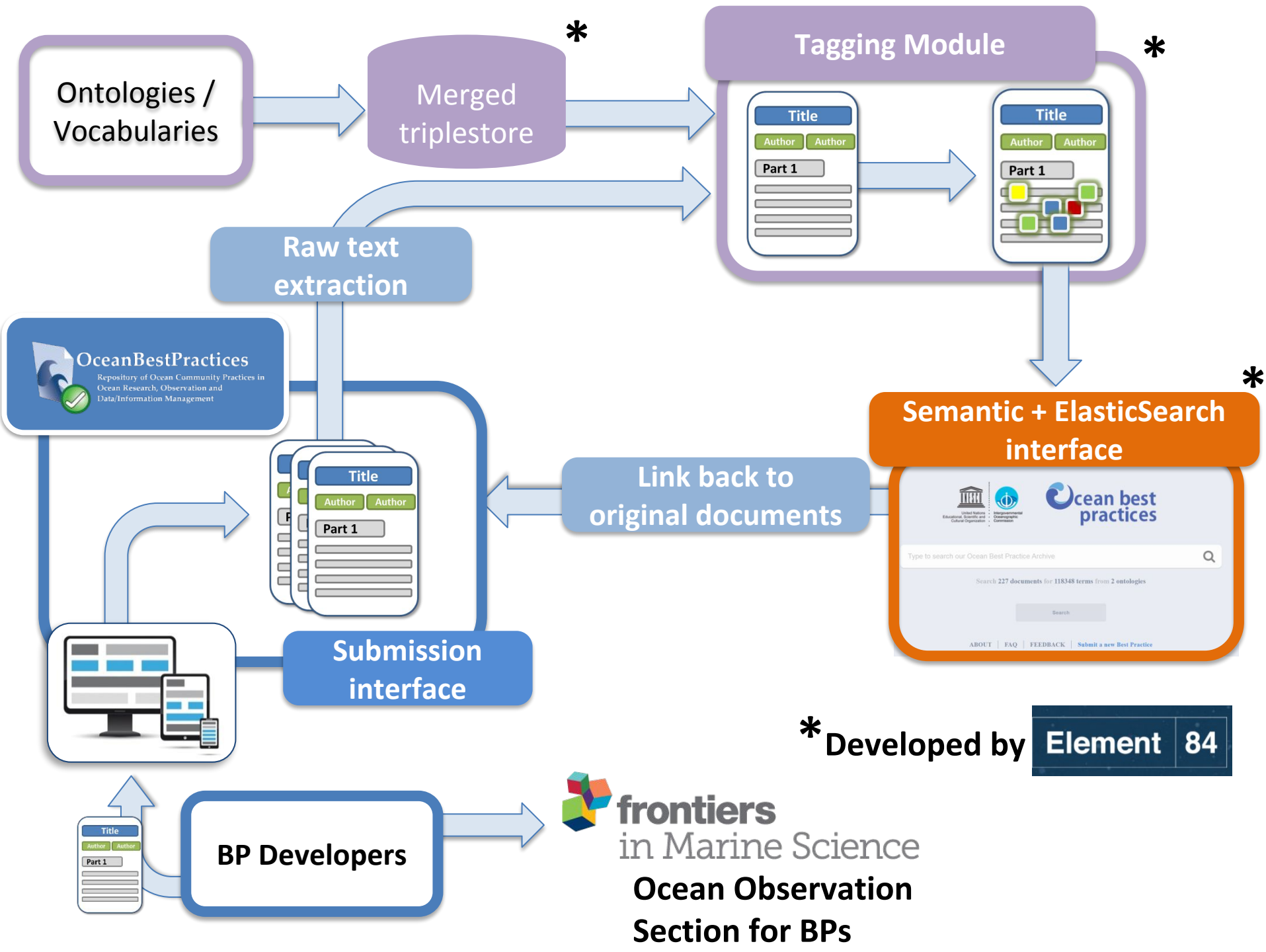


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- ⇒ [ICES : International Council for the Exploration of the Sea](#) [13]
- ⇒ [IOC : Intergovernmental Oceanographic Commission. Except IODE](#) [7]
- ⇒ [IODDE : International Oceanographic Data and Information Exchange](#) [88]
- ⇒ [IOOS: The U.S. Integrated Ocean Observing System](#) [12]
- ⇒ [JAMSTEC: Japan Agency for Marine-Earth Science and Technology](#) [6]
- ⇒ [JCOMM : Joint Tech Comm for Oceanography & Marine Meteorology](#) [35]
- ⇒ [JERICO: Joint European Research Infrastructure network for Coastal Observatories](#) [7]
- ⇒ [ODIP: Ocean Data Interoperability Platform](#) [5]
- ⇒ [SCOR: Scientific Committee on Oceanic Research](#) [2]
- ⇒ [WMO : World Meteorological Organization](#) [22]
- ⇒ [ZZZ Other Communities \(RDA; CODATA ...\)](#) [10]

**Adding semantic technology...**





# OceanBestPractices

Repository of Ocean Community Practices in  
Ocean Research, Observation and  
Data/Information Management

## BROWSE

All of OceanBestPractices

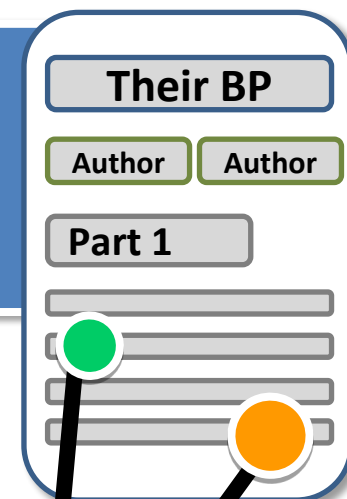
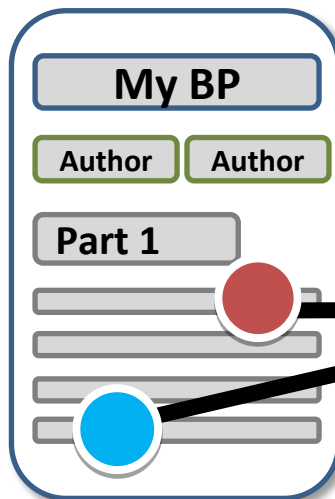
Communities & Collections

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**Sensors, Environments, Chemicals,**  
**SDG content...**



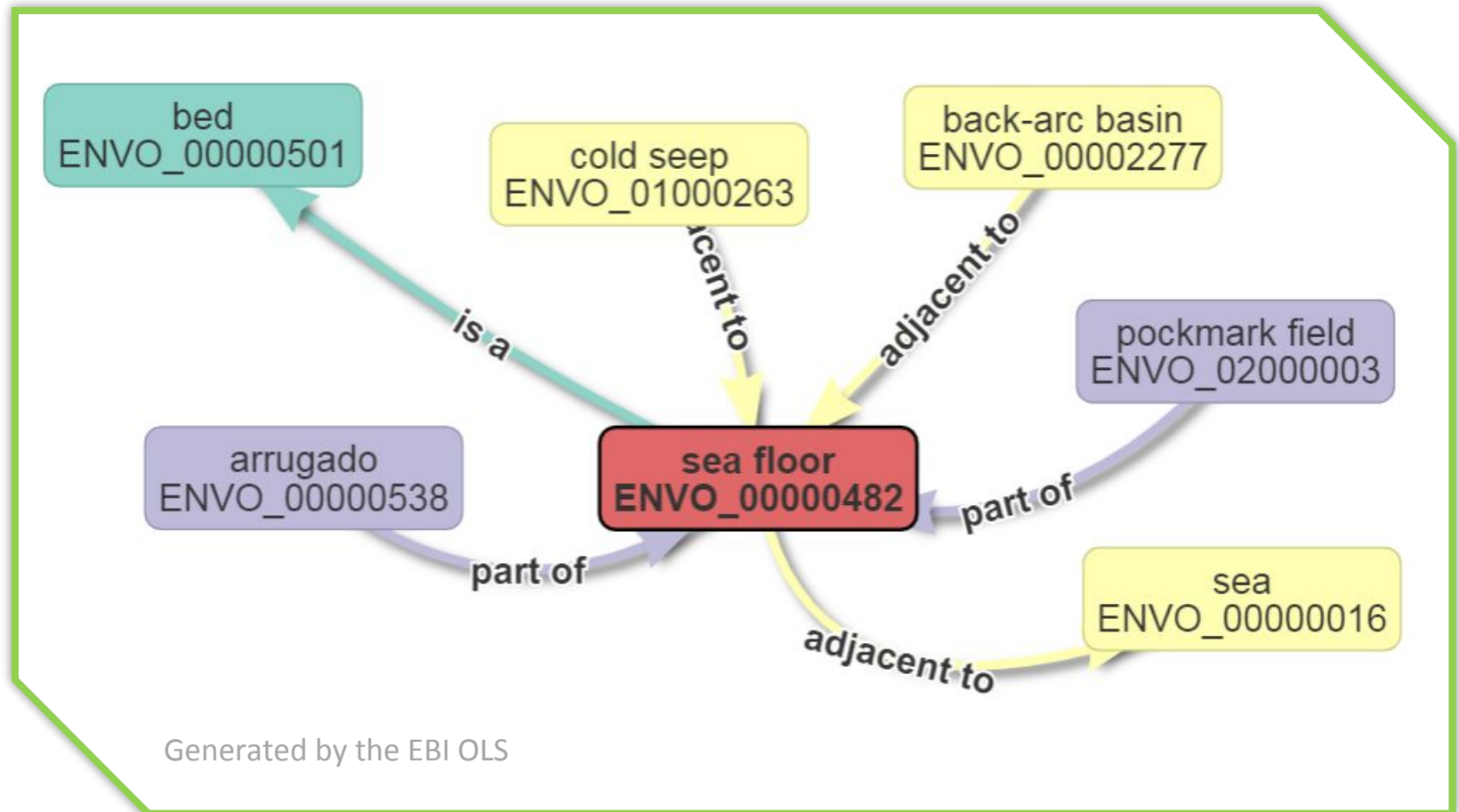
**BODC sensor vocabs**



OBO Foundry Ontologies

**ENVO, CHEBI, SDGIO**

This document includes methods to sense seepage of **methane** from the **seafloor** ...

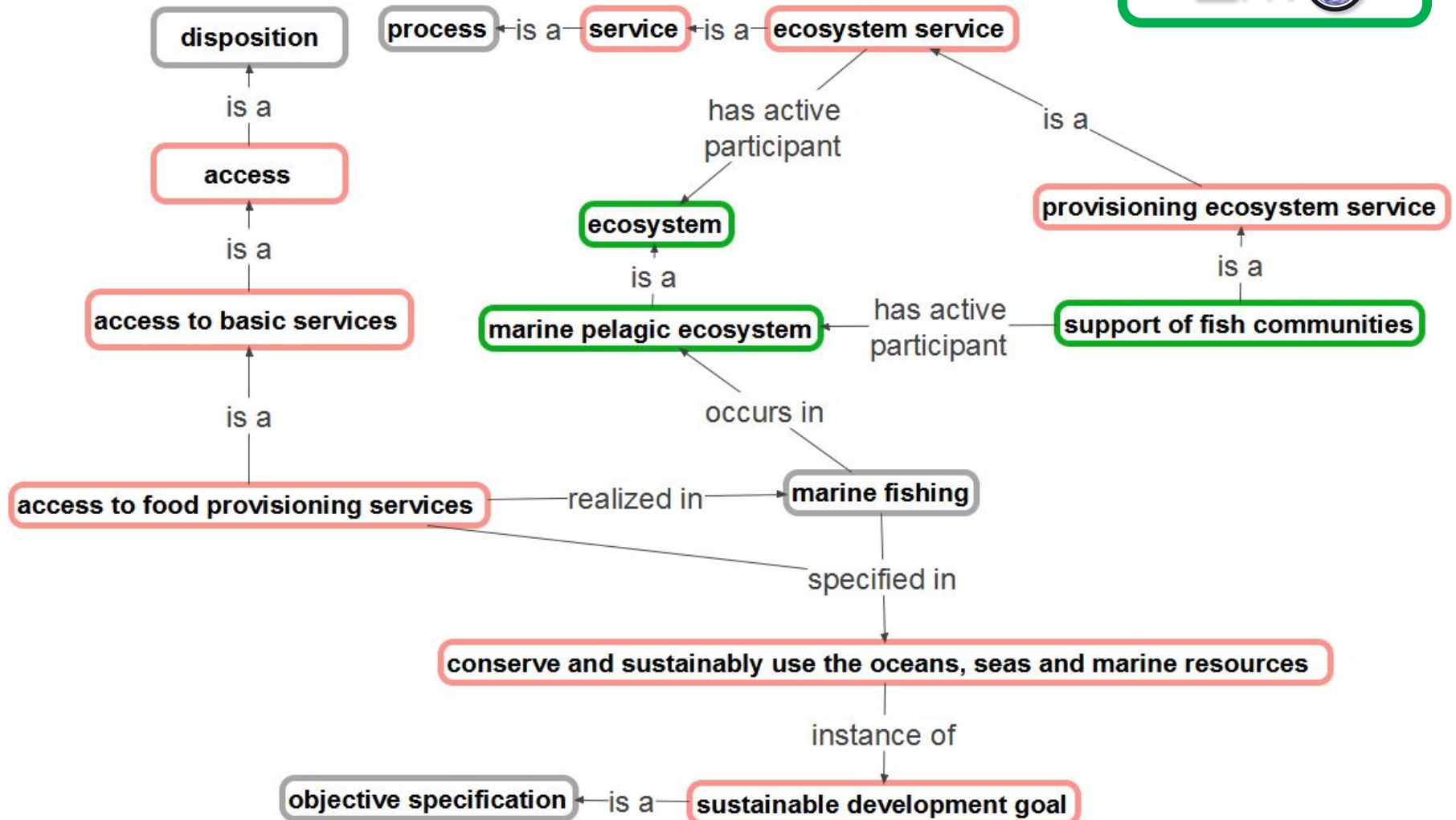




**A semantic interface for the  
Sustainable Development Goals**

**More on the ontology in this afternoon's  
SDG14/15 session**

# Linking machine knowledge between SDGs and ecosystems





# OceanBestPractices

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All of OceanBestPractices

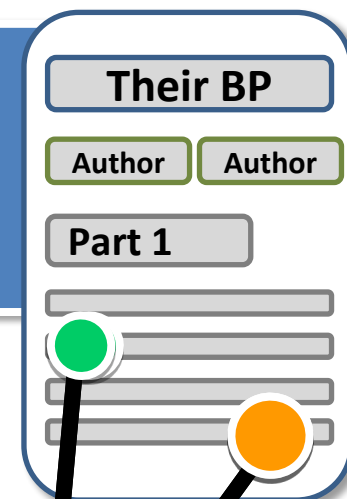
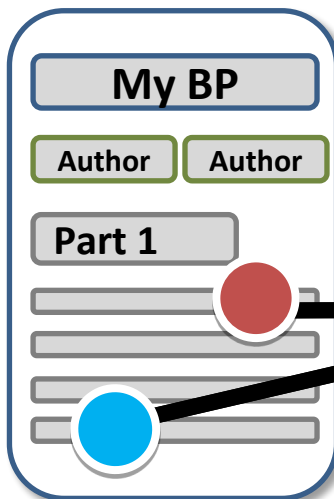
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OBO Foundry Ontologies

**ENVO, CHEBI, SDGIO**



**BODC sensor vocabs**



Type to search our Ocean Best Practice Archive



Search 227 documents for 119896 terms from 3 ontologies

Search

ENVO, CHEBI,  
SDGIO

[ABOUT](#)

[FAQ](#)

[FEEDBACK](#)

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sea ice



*sea ice*

*sea ice* formation process

*sea ice* floe

[ABOUT](#)

[FAQ](#)

[FEEDBACK](#)

[Submit a new Best Practice](#)

"sea ice"

## ORGANIZATIONS THAT SUPPORT THIS EFFORT

IODE

BPWC

### DOCUMENT TAGS

first year ice (ENVO)

sea ice (ENVO)

marine environment (ENVO)

new ice (ENVO)

frazil ice (ENVO)

pancake ice (ENVO)

ice shelf (ENVO)

Viewing 55 results for "sea ice" Page 1 of 6

2017

en

### Sea-Ice Information Services in the World. 2017 edition

A common numerical ice-reporting code (the Baltic *Sea Ice Code*) (international *sea-ice*, *sea ice* drifting., 1/10 *sea ice* coverage., *Pan* *Sea Ice* Atlas - an Arctic compendium of *sea-ice* information., *S*

< Viewing Tags

World Meteorological Organization

Explore knowledge neighbourhood  
to enhance discovery

Cryosemantics working group

"sea ice"

ORGANIZATIONS THAT SUPPORT THIS EFFORT

IODE

BPWC

Showing tags for ice shelf

is a

☐ ice mass

part of

☐ ice rise

Viewing 55 results for "sea ice" Page 1 of 6

2017

en

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A common numerical ice-reporting code (the Baltic *Sea Ice Co*  
(international *sea-ice*, *sea ice* drifting., 1/10 *sea ice* coverage., Pu  
*Sea Ice* Atlas - an Arctic compendium of *sea-ice* information., S

< Viewing Tags

World Meteorological Organization

Leveraging contextual domain  
knowledge to enhance search

# Interfacing with EuroGEOSS

- Quality data and information products and streams rest on best practices along the value chain
- EuroGEOSS marine best practices (from planning to data archiving) have a natural home in the UNESCO/IOC-IODE OBP archive and system
- The ontology classes (terms) used to tag best practices are equally suited to linking data across distributed systems
- Ontology-linked data holdings and methodology stores are a core piece of a FAIR-compliant datascape

<https://tinyurl.com/EuroGEOSS-sdgio-input>

# Links

- The OBP System
  - [www.oceanbestpractices.org](http://www.oceanbestpractices.org) (last release)
  - <https://tinyurl.com/obpBeta> (upcoming release)
  - <https://www.oceanbestpractices.net/feedback> (your feedback)
- Input for the SDG Interface Ontology
  - <https://tinyurl.com/EuroGEOSS-sdgio-input>

